

Contraindication entries : 30675
Train positive entries : 6571

Computing binary metrics ...
Baseline [val] AUROC=0.9262 AUPRC=0.9347 F1=0.8406
Baseline [test] AUROC=0.9173 AUPRC=0.9274 F1=0.8339
HGT [val] AUROC=0.9533 AUPRC=0.9577 F1=0.8824
HGT [test] AUROC=0.9453 AUPRC=0.9506 F1=0.8754

Computing ranking metrics ...
Baseline MRR=0.0586 Hits@10=0.0785 NDCG@10=0.0562
HGT MRR=0.0407 Hits@10=0.1169 NDCG@10=0.0482

Computing safety metrics ...
Baseline leakage@10=0.0318 contra@10=0.0228
HGT leakage@10=0.0454 contra@10=0.0128

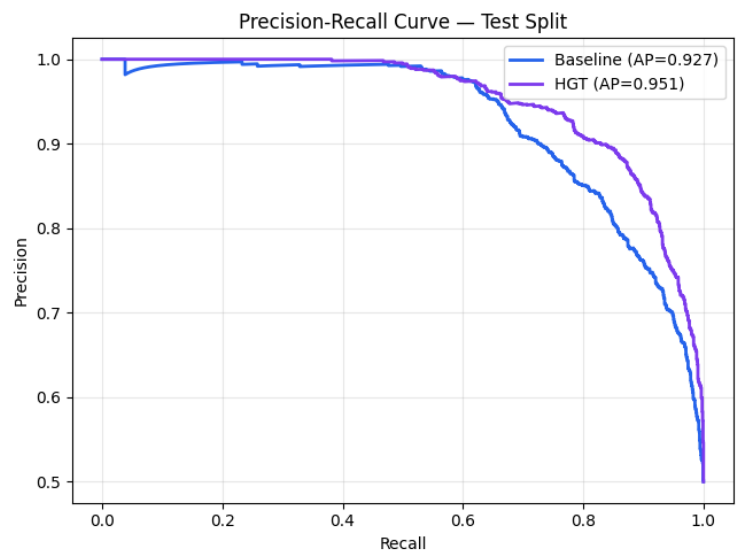
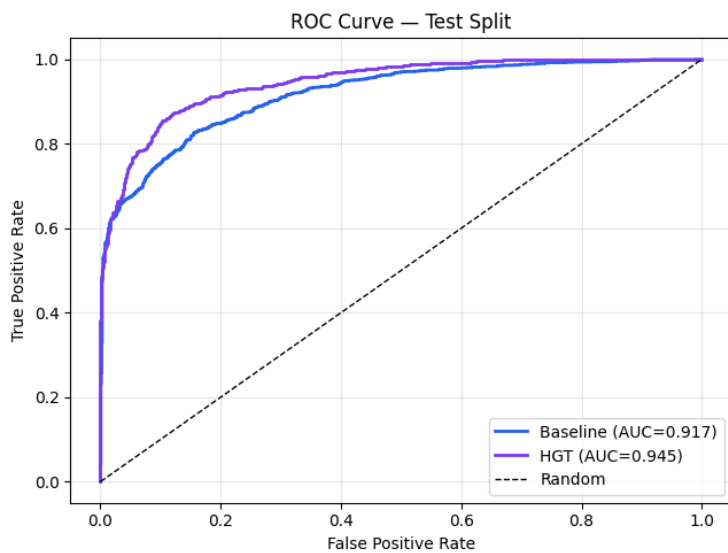
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MODEL COMPARISON — TEST SPLIT

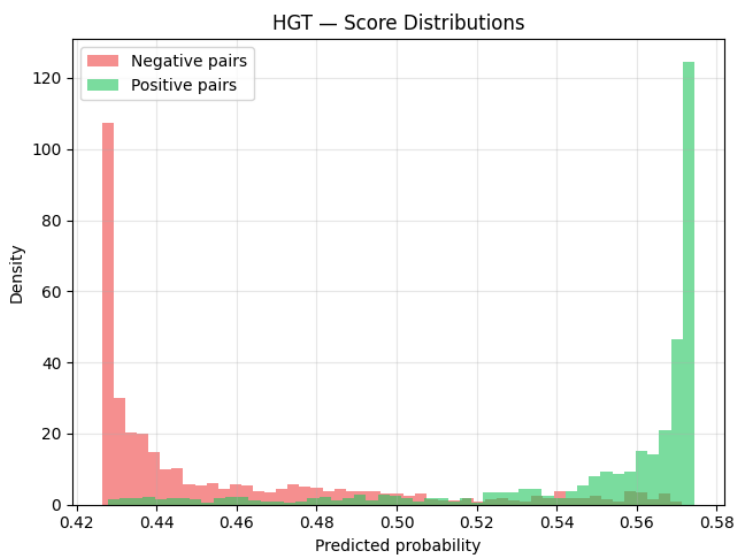
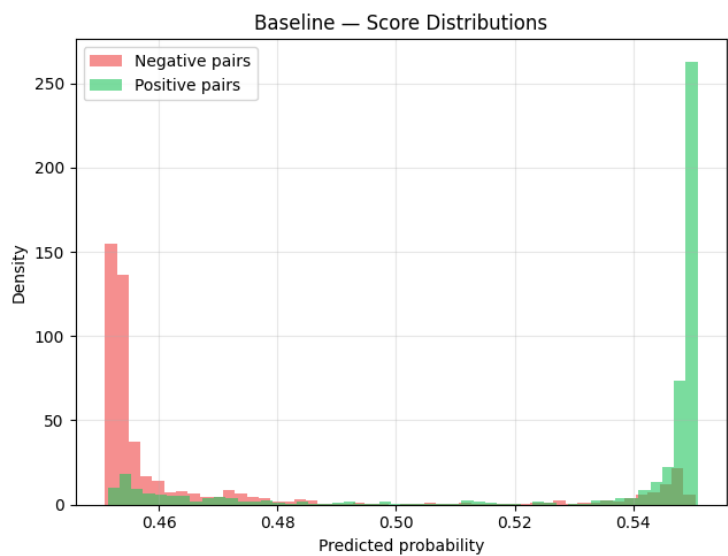
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		Baseline	HGT
Group	Metric		
Binary Classification	auroc	0.9173	0.9453
	auprc	0.9274	0.9506
	f1	0.8339	0.8754
	mcc	0.6708	0.7516
	accuracy	0.8353	0.8758
	balanced_accuracy	0.8353	0.8758
	precision	0.8412	0.8779
	recall	0.8268	0.8730
	log_loss	0.6344	0.6049
	brier_score	0.2207	0.2061
	ece	0.2879	0.3147
	pos_mean_score	0.5318	0.5512
	neg_mean_score	0.4692	0.4562
	score_gap	0.0626	0.0950
	threshold	0.4889	0.5062
Ranking	mrr	0.0586	0.0407
	mean_rank	810.8956	773.9478
	median_rank	436.0000	345.0000

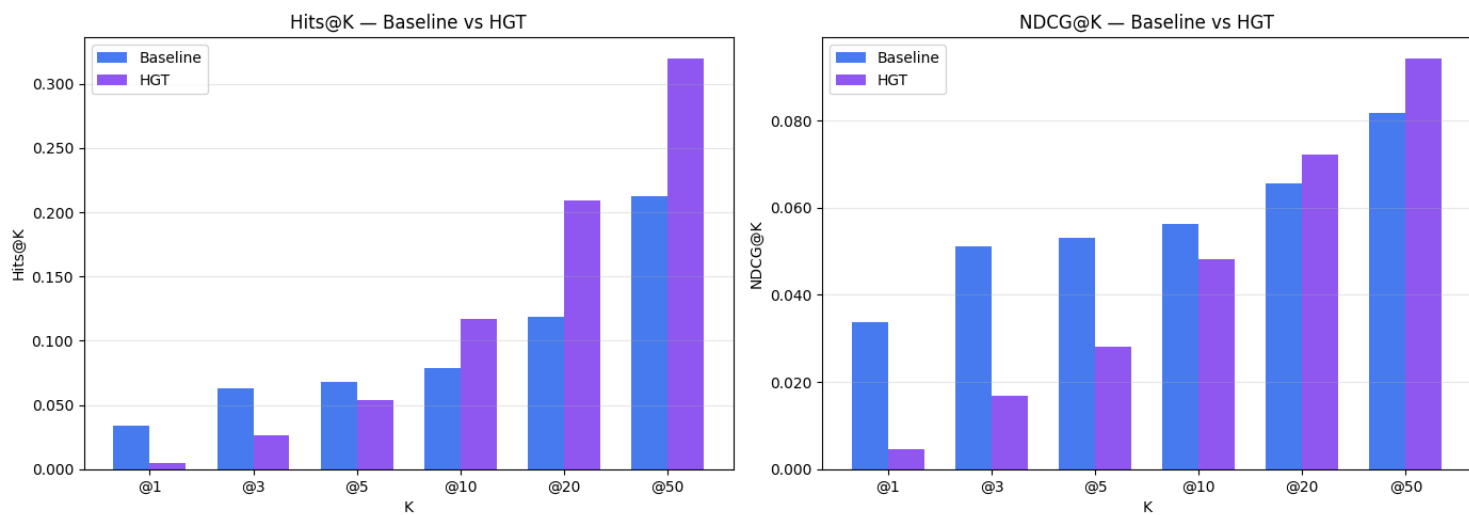
	hits@1	0.0338	0.0046
	hits@3	0.0631	0.0262
	hits@5	0.0677	0.0538
	hits@10	0.0785	0.1169
	hits@20	0.1185	0.2092
	hits@50	0.2123	0.3200
	recall@10	0.0362	0.0669
	recall@50	0.1211	0.2240
	precision@10	0.0083	0.0128
	ndcg@10	0.0562	0.0482
	ndcg@50	0.0817	0.0943
Safety	leakage_rate@10	0.0318	0.0454
	contra_rate@10	0.0228	0.0128



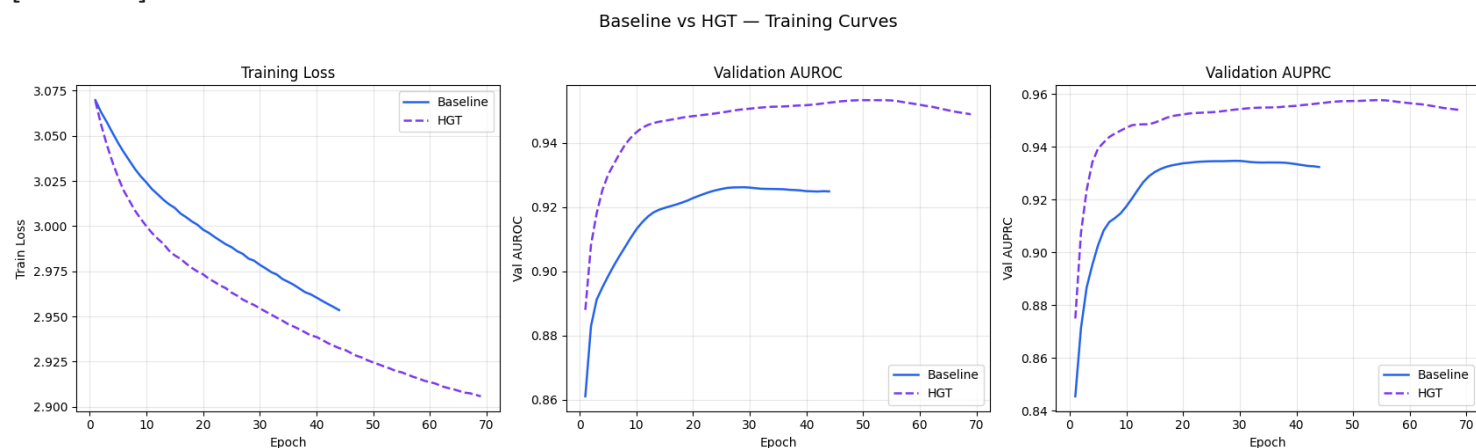
[Plot 1+2] ROC and PR curves saved.



[Plot 3] Score distribution histograms saved.



[Plot 4] Hits@K and NDCG@K bar charts saved.



[Plot 5] Training curves overlay saved.

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BLOCK 09 COMPLETE

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Objects now available:

```
binary_metrics      {model_name: {split: metrics_dict}}
ranking_metrics     {model_name: metrics_dict}
safety_metrics      {model_name: metrics_dict}
comparison_df       full side-by-side DataFrame
```

Figures saved to /mnt/data/:

```
eval_roc_pr_curves.png
eval_score_distributions.png
eval_ranking_bars.png
eval_training_curves_overlay.png
```

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